NaturArv3D Compression Study

Meeting 2026-08-12

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Introduction

- About me: Jeppe
  - MSc student in Mathematical Modelling and Computation at DTU
  - R&D Scientist in NMR sensor technology company
  - BA (MA) (Hons) Natural Science — Physics at Cambridge
- Started in QIM Center in July 2026
- Assigned to NaturArv3D project
  - Reachable at `s250250@dtu.dk`

Motivation

- NaturArv3D will generate **lots** of data
  - Proj[1], 4(**2**)K@16bit: ≈ 50(**13**)GB
  - Recon, 4K@32(**16**)bit: = 128(**64**)GB
  - Recon, 2K@32(**16**)bit: = 16(**8**)GB
  - × 500 samples/week ⟹ 10-50 TB/week!
  - ⟹ 1500 — 7500 DKK per week[2]? ~100 kDKK per year?
- **Finding correct compression scheme enables significant cost savings**
- No up-to-date, comprehensive studies available in the literature
- [1], 2015: storage cost significant portion of $3.5 billion rollout of Electronic Health Records in Canada

[1] Assumes 1572 projections (which seems ambitious given 500 samples/week)

[2] Hand-wavey estimate of 150 DKK/TB for 5 years of redundant HDD storage

Current Goals

Phase 0 (today, later slide?)

- Determine requirements around formats, compatibility, quality, speed, etc. for reconstructions (NHMD, QIM?), and projections (QIM)
  - TIFF-stack required, or Zarr acceptable for reconstructions?
  - How exotic of a codec can we use?

Phase 1 (soon)

- Study compression ratios (CR) and encoding/decoding speeds for different **lossless** codecs on **projection** *and maybe reconstruction* data
- Study CRs, speeds, and *quality* for different **lossy** codecs on **reconstruction** data

Phase 2 (optional?)

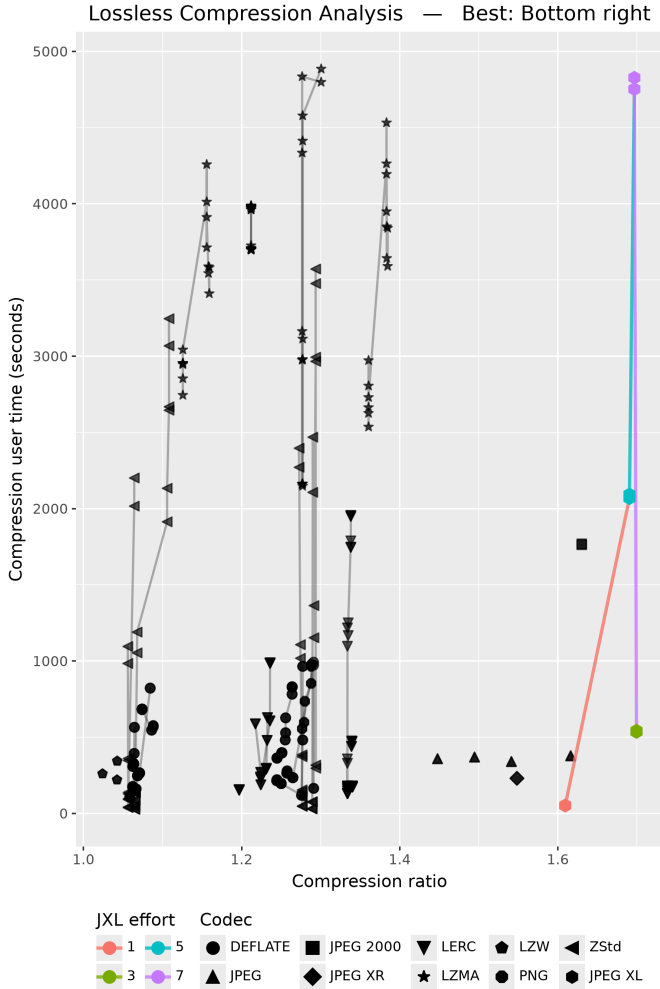
- Study effect of **lossy** compression of **projection** data on reconstruction quality for FDK, maybe iterative?

Literature Review

- Literature suggests [2], [3]:
  - Projections: $\mathrm{CR} \in [1.7, 2]$ for lossless
  - Projections: $\mathrm{CR} \approx 3$ for near-lossless (indisguishable reconstructions)
  - Reconstructions: National Medical Boards: $\mathrm{CR} \in [5, 30]$ (lossy) [4]
- Recent JPEG-XL codec adopted by DICOM standard [5]
- LERC codec allows for guaranteed error bounds
  - Good candidate for lossy compression of projections

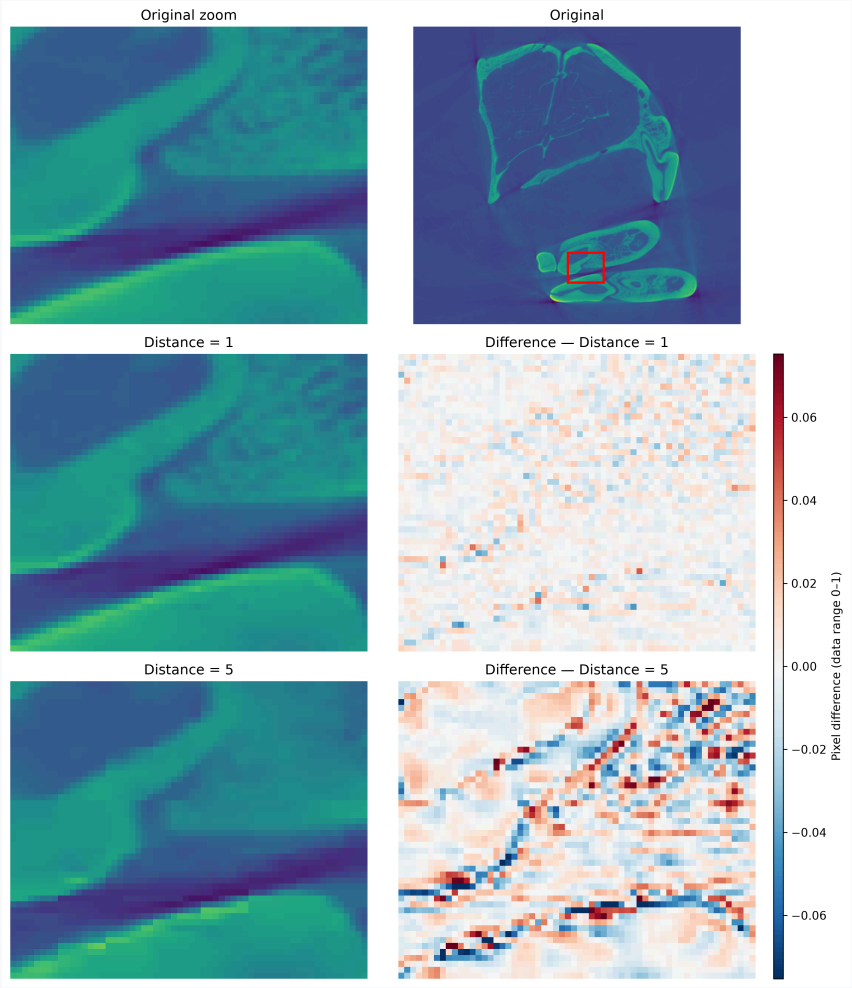
Preliminary Experiments: Lossless

- Lossless compression of projections
  - As a rule, 50% space saving is best case for 16-bit XCT data [2], [6]
- Across 12 codecs and 506 configurations:
  - JPEG-XL (v0.12) performs best
    - CR ≈ 1.7 ⟹ 41% space saving
  - JPEG-2K, JPEG-LS close (CR ≈ 1.62)
  - PNG, ZStd, LERC, DEFLATE, … much worse (CR ≈ 1.3)
- JPEG-XL pros: u8/u16/f16/f32 support, tunable, high CR, open-source, adopted by DICOM
- JPEG-XL cons: very new, immature ecosystem, no support in older software



Preliminary Experiments: Lossy

- 3D reconstructions take up bulk of storage
- Can be remade from archived projections
- This enables lossy compression
- Lossy JPEG-XL:
  - Distance 1 (middle row) gives 98.9% space saving
    - CR=90.5, SSIM=0.9992
  - Distances >1 produce noticeable artefacts and smoothening
  - Distance 0.1 gives 92.1% space saving
    - CR=12.7, SSIM=0.9995
- No guaranteed error bounds in JPEG-XL
  - Future experiments include LERC for error bounds



Discussion: Scope

- NHMD requirements and desired output?
- NHMD wants only *reconstructions* (?)
  - Desired/required format?
  - Note: TIFF is container format ⟹ *a TIFF is not just a TIFF*
    - *Many* compression schemes available, **only few with wide-spread support**
      - DEFLATE, LZW — both lossless, bad CR
- Help me understand use-cases
  - How will data be used?
- Archivability concerns
  - Offer: Produce x86_64 Linux container image with decoder
    - Should be runnable also in 50 years
- QIM/3DIM/DTU wants *projections*, maybe reconstructions
  - Wants to store as close to lossless as possible

Bibliography

[1] J.-F. Pambrun and R. Noumeir, “Computed Tomography Image Compressibility and Limitations of Compression Ratio-Based Guidelines,” *Journal of Digital Imaging*, vol. 28, no. 6, pp. 636–645, Dec. 2015, doi: 10.1007/s10278-015-9791-7.

[2] A. Wegener, N. Chandra, Y. Ling, R. Senzig, and R. Herfkens, “Real-Time Compression of Raw Computed Tomography Data: Technology, Architecture, and Benefits,” presented at the SPIE Medical Imaging, E. Samei and J. Hsieh, Eds., Lake Buena Vista, FL, Feb. 2009, p. 72582H. doi: 10.1117/12.810599.

[3] A. Wegener, N. Chandra, Y. Ling, R. Senzig, and A. Et., “Effects of Fixed-Rate CT Projection Data Compression on Perceived and Measured CT Image Quality,” *SPIE Proceedings*, 2010, doi: 10.1117/12.841145.

Bibliography

[4] F. Liu, M. Hernandez-Cabronero, V. Sanchez, M. W. Marcellin, and A. Bilgin, “The Current Role of Image Compression Standards in Medical Imaging,” *Information (Basel)*, vol. 8, no. 4, p. 131, Dec. 2017, doi: 10.3390/info8040131.

[5] “DICOM PS3.5 2026c — Data Structures and Encoding.” Accessed: Jul. 17, 2026. [Online]. Available: https://www.dicomstandard.org/current

[6] F. Marone, J. Vogel, and M. Stampanoni, “Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices,” *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.